IV Bolus Practice 1

Module 2 - IV Bolus Administration · Daily practice sheet · No date

Name _____ Date _____

a The data below are plasma concentrations of a new drug at different time points after an intravenous bolus injection of 50 mg to a 70-kg patient. Time (hr) and plasma concentration (mg/L): 0.5 and 2.52; 1.0 and 1.59; 1.5 and 1.00; 2.0 and 0.64; 2.5 and 0.40; 3.0 and 0.25. Using the 1.0 hr and 2.0 hr points, what is the elimination rate constant?

b A drug given by intravenous bolus injection has an elimination rate constant of 0.91 hr^{-1}. What is its half-life of elimination?

c After an intravenous bolus injection of 50 mg, a drug with an elimination rate constant of 0.91 hr^{-1} has a plasma concentration of 0.40 mg/L at 2.5 hours. What is the initial plasma concentration?

d A drug given by intravenous bolus injection has an initial plasma concentration of 3.89 mg/L and an elimination rate constant of 0.91 hr^{-1}. What is the concentration of drug in the plasma 15 minutes after the dose was given?

e A 50 mg intravenous bolus dose produces an initial plasma concentration of 3.89 mg/L in a 70-kg patient. What is the apparent volume of distribution?

f A drug has an apparent volume of distribution of 12.85 L and an elimination rate constant of 0.91 hr^{-1} in a patient. What is the total body clearance of this drug in this patient?

g A 50 mg intravenous bolus dose of a drug with a volume of distribution of 12.85 L gives a plasma concentration of 0.25 mg/L at 3 hours. What is the amount of drug in the body 3 hours after the drug was administered?

h A drug given by intravenous bolus injection has an elimination half-life of 0.76 hours. How much time is required for 99.9% of the drug to be eliminated from the body?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 0.91 hr^{-1}

WHAT THE STEM GIVES

D_0	50 mg	*not needed: k comes from two concentrations alone*
weight	70-kg	*not needed*
C_{p^1}	1.59	the 1.0 hr concentration, in mg/L, the numerator of the ratio
t_1	1.0	the earlier time, in hr
C_{p^2}	0.64	the 2.0 hr concentration, the denominator of the ratio
t_2	2.0	the later time; $t_2 - t_1 = 1.0$ hr
other points	2.52, 1.00, 0.40, 0.25	*not needed: the stem names the 1.0 and 2.0 hr pair*

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = -\text{slope} = \frac{\ln Cp1 - \ln Cp2}{t_2 - t_1}$$, taking the 1.0 hr and 2.0 hr points, which is the pair Dr. Mosley works in her own key

On semi-logarithmic axes a first-order decline is a straight line with slope −k, so ln of two concentrations over the elapsed time gives k. Her key uses this pair; the concentrations are rounded to two decimals, so other pairs differ slightly. k is never negative, so the minus sign goes on the slope.

ALGEBRA

$$k = \frac{[\ln 1.59 - \ln 0.64]}{(2.0 - 1.0)\ \text{hr}} = \frac{\ln(1.59/0.64)}{1.0\ \text{hr}}$$

The difference of two natural logarithms is the natural logarithm of their ratio, which is the rearrangement she uses because the concentration units cancel inside the ratio and leave a pure number. Dividing that pure number by a time in hours is what produces reciprocal hours.

ALGEBRA

$$k = \frac{0.91002}{1.0\ \text{hr}} = 0.91002\ hr^{-1}$$

The ratio $\frac{1.59}{0.64}$ is 2.4844 and its natural logarithm is 0.91002. Because the two sampling times are exactly one hour apart the division leaves the logarithm unchanged in magnitude and only attaches the reciprocal-hour unit.

ROUND

$k = 0.9100\ hr^{-1}$

Report rate constants to four decimal places. A k rounded early to one or two figures carries a large error into every later part of the problem, and four decimals is the precision her key prints for this one.

DOES THE NUMBER MAKE SENSE?

From 1.0 to 2.0 hr the concentration falls 1.59 → 0.64 mg/L, more than half in one hour, so the half-life is under 1 hr and k is above 0.693 hr^{-1}; 0.91 fits.

Other neighbouring pairs of points in this table give k values from 0.893 to 0.940 hr^{-1}. All of them are accepted: the spread comes from rounding in the printed concentrations, not from a mistake in method.

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b

Her key

Answer: 0.76 hr

WHAT THE STEM GIVES

k 0.91 hr^{-1} the denominator of $\frac{0.693}{k}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½ = \frac{0.693}{k}$$

Setting the concentration to half its starting value in $C_p = C_p^{0} \cdot e^{-kt}$ leaves the natural logarithm of two, 0.693, divided by the rate constant. This relationship is not on the equation sheet, so it has to be recalled rather than looked up.

ALGEBRA

$$t½ = \frac{0.693}{0.91\ hr^{-1}}$$

The rate constant sits in the denominator of the rearranged equation. Dividing a pure number by a quantity in reciprocal hours gives hours, which is the check that the answer will come out as a time rather than a rate.

ROUND

$t½ = 0.76\ hr$

The division gives 0.7615, which her key prints as 0.76 hr, two decimal places. Half-life must be reported in units of time and never in reciprocal time, a distinction she calls out by name in the exam review.

DOES THE NUMBER MAKE SENSE?

A k above 0.693 hr^{-1} means the half-life is under 1 hr, and 0.693 over 0.91 is about three quarters of an hour; 0.76 hr fits.

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c

Her key

Answer: 3.89 mg/L

WHAT THE STEM GIVES

D_0	50 mg	*not needed: the decline is worked in concentration*
k	0.91 hr^{-1}	the exponent kt = 0.91 × 2.5
C_p	0.40 mg/L	the sampled concentration, multiplied by e^{kt} = 9.7279
t	2.5 hours	the sample time in the exponent

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p^0 = C_p \cdot e^{+kt}$$

Going backwards along the curve from a measured point to time zero reverses the sign of the exponent in $C_p = C_p^0 \cdot e^{-kt}$. Keeping the minus sign gives a number lower than the one you started from, which cannot be right: the initial concentration must be the highest point in the data set.

ALGEBRA

$$C_p^0 = (0.40\ mg/L) \cdot e^{0.91\ hr^{-1} \times 2.5\ hr}$$

Reciprocal hours multiplied by hours leaves the exponent dimensionless, which it must be. The 2.5 hr is the full elapsed time from the dose to the sample being extrapolated from, so the whole decline over that interval is undone in one step.

ALGEBRA

$$C_p^0 = (0.40\ mg/L)(9.7279) = 3.8912\ mg/L$$

The exponent is 2.275 and e raised to it is 9.7279. Multiplying a concentration by a pure number keeps the concentration unit, so the answer is in milligrams per litre.

ROUND

$$C_p^0 = 3.89\ mg/L$$

Her key prints 3.89 mg/L, three significant figures. The value is higher than the earliest measured concentration of 2.52 mg/L at 0.5 hr, which is the sanity check she asks for: the initial concentration should be the highest point.

> **DOES THE NUMBER MAKE SENSE?**
> With k = 0.91 hr^{-1} the half-life is about three quarters of an hour, so 2.5 hours is more than three half-lives: C_p^0 is more than eight times 0.40 mg/L, and 3.89 mg/L fits.

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d

Her key

Answer: 3.1 mg/L

WHAT THE STEM GIVES

C_p^0	3.89 mg/L	the starting concentration, multiplied by e^{-kt}
k	0.91 hr^{-1}	the exponent kt = 0.91 × 0.25
t	15 minutes	becomes 0.25 hr so the exponent is a pure number

THE WORKING, ONE LINE AT A TIME

UNIT

$$15\ \text{min} \times \frac{1\ \text{hr}}{60\ \text{min}} = 0.25\ \text{hr}$$

The rate constant is in reciprocal hours, so the time must be in hours before the two are multiplied or the exponent will be sixty times too large. She converts 15 minutes to 0.25 hr silently in her key, but the conversion is the step that decides whether the answer is right.

SETUP

$$C_p = C_p^{\,0} \cdot e^{-kt}$$

This is the forward form of the integrated first-order equation, used whenever a concentration at a stated later time is wanted from a known starting concentration. The minus sign belongs here because the concentration is falling.

ALGEBRA

$$C_p = (3.89\ \text{mg/L}) \cdot e^{-0.91\ \text{hr}^{-1} \times 0.25\ \text{hr}} = (3.89\ \text{mg/L})(0.79652)$$

The exponent is −0.2275, and e raised to it is 0.79652, the fraction of the starting concentration still present. Multiplying the initial concentration by that fraction gives the concentration at 15 minutes.

ROUND

$$C_p = 3.0985\ \text{mg/L, reported as } 3.1\ \text{mg/L}$$

Her key prints 3.1 mg/L. Since a quarter of an hour is well under one half-life of 0.76 hr, the answer must lie between 3.89 and half of 3.89, which it does.

> **DOES THE NUMBER MAKE SENSE?**
> A quarter hour is about a third of the half-life of three quarters of an hour, so only about 20 per cent is lost: C_p sits a little under 3.89 mg/L and above half of it, and 3.1 mg/L fits.

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e

Her key

Answer: 12.85 L

WHAT THE STEM GIVES

D_0	50 mg	the numerator: the amount in the body at time zero
$C_p^{\,0}$	3.89 mg/L	the denominator
weight	70-kg	*not needed: V_D comes from the dose and the concentration*

THE WORKING, ONE LINE AT A TIME

SETUP

$$V_D = \frac{D_B}{C_p}, \text{ with } D_B = D_0 \text{ at time zero}$$

The volume of distribution is the proportionality constant between the amount of drug in the body and the measured plasma concentration, so rearranging $C_p = \frac{D_B}{V_D}$ puts the amount on top. Immediately after a bolus the amount in the body is the whole dose, which is what lets the dose be used here.

ALGEBRA

$$V_D = \frac{50 \text{ mg}}{3.89 \text{ mg/L}}$$

The milligrams cancel between numerator and denominator and the litre in the denominator of the concentration inverts, leaving litres. That unit check is also the apples-to-apples check she insists on: an amount is divided by a concentration, never by another amount.

ROUND

$V_D = 12.853$ L, reported as 12.85 L

Her key prints 12.85 L, two decimal places. The patient weight of 70 kg is given in the stem and is not used, which is a habit of hers in these problems.

> **DOES THE NUMBER MAKE SENSE?**
> 50 mg over 4 mg/L would be near 12 L, and 3.89 is a little under 4, so V_D is a little over 12 L; 12.85 L fits, roughly a fifth of the 70-kg body weight.

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f

Her key

Answer: 11.7 L/hr

WHAT THE STEM GIVES

V_D	12.85 L	one factor of the product $k \cdot V_D$
k	0.91 hr^{-1}	the other factor

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = k \cdot V_D$$

Total body clearance is the product of the overall elimination rate constant and the volume of distribution. This relationship is one of the two she says will not be supplied on the equation sheet, so it has to be recalled. Both factors are constants, which is why their product is constant too.

ALGEBRA

$$Cl_T = (12.85 \text{ L})(0.91 \text{ hr}^{-1}) = 11.6935 \text{ L/hr}$$

Litres multiplied by reciprocal hours gives litres per hour, which is the unit a clearance must carry. A result in plain litres would mean the rate constant had been left out and a volume of distribution reported instead.

ROUND

$Cl_T = 11.694$ L/hr, reported as 11.7 L/hr

Her key prints 11.7 L/hr, three significant figures. Carrying both inputs unrounded, as V_D = 12.8535 L and k = 0.91002 hr^{-1}, gives 11.697 L/hr, which rounds to the same value.

> **DOES THE NUMBER MAKE SENSE?**
> 0.91 is a little under 1, so $k \times V_D$ is a little under 12.85 L/hr; 11.7 L/hr fits.

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g

Her key

Answer: 3.2 mg

WHAT THE STEM GIVES

D_0	50 mg	*not needed on the main route: the amount at 3 hr is $V_D \cdot C_p$*
V_D	12.85 L	one factor of $D_B = V_D \cdot C_p$
C_p	0.25 mg/L	the 3-hour concentration, the other factor
t	3 hours	*not needed on the main route; it enters only the e^{-kt} check*

THE WORKING, ONE LINE AT A TIME

SETUP

$D_B = V_D \cdot C_p$, evaluated at 3 hr

The volume of distribution converts a concentration into an amount at any time, not only at time zero, because it is a constant. Since the plasma concentration at 3 hours is already given, this is the shorter of the two routes her key shows.

ALGEBRA

$D_B = (12.85\ \text{L})(0.25\ \text{mg/L}) = 3.2125\ \text{mg}$

Litres multiplied by milligrams per litre leaves milligrams, which is the unit an amount must have. Getting litres out instead would mean the two quantities had been divided rather than multiplied.

ALGEBRA

Alternative route: $D_B = D_0 \cdot e^{-kt} = (50\ \text{mg}) \cdot e^{-0.91\ \text{hr}^{-1} \times 3\ \text{hr}} = 3.2610\ \text{mg}$

The same amount can be found by decaying the whole dose forward for three hours, which avoids needing the volume of distribution. She prints both routes deliberately. The two differ slightly only because the 0.25 mg/L reading is rounded to two decimal places in the data table.

ROUND

$D_B = 3.2$ mg by either route

Her key gives 3.2 mg from both calculations. Reporting to two significant figures is what makes the two routes agree, which is her reason for showing them side by side.

> **DOES THE NUMBER MAKE SENSE?**
> Amount is volume times concentration: 12.85 L × 0.25 mg/L is a quarter of 12.85, about 3.2 mg; it must be far below the 50 mg given, after 3 hours of a sub-hour half-life.

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h

Her key

Answer: 7.6 hr

WHAT THE STEM GIVES

t½	0.76 hours	multiplied by the 10 half-lives that 99.9% elimination takes
fraction	99.9%	eliminated, which leaves 0.1%, one part in 1000: 10 half-lives

THE WORKING, ONE LINE AT A TIME

SETUP

99.9% eliminated corresponds to 10 half-lives

For a first-order process each half-life leaves half of what was there, so after ten half-lives 0.1% remains and 99.9% has gone. Dr. Mosley states the ten-half-life figure outright as a fact to carry, and her Module 1 key builds the same result from a table of percent remaining.

ALGEBRA

$t = 10 \times t_{½} = 10 \times 0.76 \text{ hr} = 7.6 \text{ hr}$

Multiplying a count of half-lives by the length of one half-life gives a time. Nothing needs to be exponentiated because the percentage asked for is a power of one half, which is the route she prefers whenever it is available.

ALGEBRA

Logarithmic route: $t = \frac{\ln(100/0.1)}{k} = \frac{\ln(1000)}{0.91 \text{ hr}^{-1}} = 7.591 \text{ hr}$

Setting the fraction remaining to 0.001 in $C_p = C_p^{\,0} \cdot e^{-kt}$ and solving for t gives the natural logarithm of 1000 divided by k. This is the general route for percentages that are not powers of one half, and she prints it beside the half-life count in the parallel problem.

ROUND

$t = 7.6 \text{ hr}$

Her key gives 7.6 hr. The small gap between 7.6 and the exact 7.59 is the rounding already carried in the half-life of 0.76 hr, so both routes report the same answer at this precision.

DOES THE NUMBER MAKE SENSE?

99.9% eliminated leaves 0.1%, one part in 1000, which is ten halvings, so $t = 10 \times 0.76 \text{ hr} = 7.6 \text{ hr}$.